

Question#1 Number of Faculty: The numbers of faculty at 32 randomly selected state-controlled colleges and universities with enrollment under 12,000 students are shown below.

211 384 396 211 224 337 395 121 356 621 367 408 515 280 289 180 431 176 318 836 203 374
224 121 412 134 539 471 638 425 159 324

- a) Use the data and estimate the population mean
- b) Find the unbiased estimator of the population variance
- c) Find the biased estimator of the population variance
- d) Construct 95% Confidence interval for population Mean assuming pop variance 165.1
- e) Construct 95% Confidence interval for population Mean assuming pop variance unknown (taking critical value = 2.355)

Question#2 The annual revenues (in \$1,000s) of the Finance Departments from 32 randomly selected mid-sized companies (each with total staff under 3,000) are shown below:

42 55 61 47 38 72 69 51 58 64 76 83 95 47 39 56 74 91 68
59 73 48 62 41 84 79 53 70 58 66 63 45

- a) Use the data and estimate the population mean annual revenue (in \$1,000s).
- b) Find the unbiased estimator of the population variance.
- c) Find the biased estimator of the population variance.
- d) Construct a 95% confidence interval for the population mean, assuming the population variance = 49.0 (known).
- e) Construct a 95% confidence interval for the population mean, assuming the population variance is unknown (take the critical value = 2.042).

Question#3 A company received a revised version of its financial data showing the annual profits (in Rs. 000) from 25 randomly selected regional branches as follows:

112 135 128 149 121 138 142 156 130 117 165 158 140 133 119
144 151 124 136 155 127 148 134 120 139

- a) Find the point estimate for the population mean.
- b) Find the unbiased point estimate for the population variance.
- c) Find the biased point estimate for the population variance.

Question#5 An e-commerce firm claims that the average delivery time for express orders is less than 3.5 days. To test this claim, a researcher collects data on 45 deliveries and finds the average delivery time to be 3.7 days, with a sample standard deviation of 0.8 days. Assume the population variance is 0.64 days^2 .

Using a 90% confidence level, test the firm's claim about average delivery time. What conclusion can you draw?

Question#6 A credit card company claims that the average monthly spending by its premium customers is at least Rs.18,000. A random sample of 50 premium customers shows an average monthly expenditure of Rs.17,500, with a sample standard deviation of Rs.2,100. Assume the population variance is unknown. At a 99% confidence level, test whether the company's claim is justified. What conclusions can you draw?

Question#7 A private bank claims that the average time to approve a personal loan is no more than 5 working days. To verify this claim, a financial analyst samples 36 loan applications and finds the average approval time to be 5.4 days, with a sample standard deviation of 1.2 days.

From historical data, it is known that the population variance of approval times is 1.44 days^2 . Assuming a 95% confidence level, test the bank's claim about the average loan processing time. What conclusion can you draw?

Question#8 A retail store claims that at least 85% of its customers are satisfied with its service.

To verify this claim, a researcher surveys 200 customers, and 162 report being satisfied.

- a) Test the store's claim at a 5% level of significance.
- b) Construct a 95% confidence interval for the true proportion of satisfied customers.
- c) Based on the interval, what can you conclude about the store's claim?

Question#9 A marketing firm believes that 60% of urban consumers prefer online shopping to in-store shopping.

In a recent survey of 150 consumers, 102 indicated they prefer online shopping.

- a) Test whether the true proportion of consumers who prefer online shopping is at least 60% at a 1% significance level.
- b) Construct a 99% confidence interval for the population proportion of consumers

preferring online shopping.

Question#10 An international executive wants to know whether the average annual cost of supporting her family of four in City A is *no more than* the cost in City B . She collects independent random samples of 12 families from each city. Assume annual costs are normally distributed and population variances are equal. Use $\alpha = 0.01$.

Data (annual cost in \$):

City A: 7100, 6950, 6800, 7000, 6900, 7150, 6850, 7050, 6980, 6920, 7090, 6880

City B: 7200, 7250, 7100, 7300, 7180, 7260, 7120, 7210, 7190, 7310, 7150, 7230

- a) State the null and alternative hypotheses for an *at most* test and perform the test at $\alpha = 0.01$.
- b) Report the conclusion in practical terms.
- c) Construct a 95% confidence interval for the difference and interpret it.

Question#11 A researcher claims that City X costs *at least* as much as City Y to maintain a given lifestyle She samples 10 families from City X and 10 families from City Y. Assume normality and equal variances. Use $\alpha = 0.05$.

Data (annual cost in Rs. 000):

City X: 82, 79, 85, 81, 88, 76, 84, 80, 83, 77

City Y: 78, 74, 76, 77, 73, 75, 79, 72, 80, 74

- a) Formulate the hypotheses for an *at least* (one-sided) test and carry out the test at $\alpha = 0.05$.
- b) State the conclusion and its business implication.
- c) Construct a 90% confidence interval for the mean cost in City X and interpret.

Question#12

A family-support cost standard for comfortable living is Rs. 1,200,000 per year. A planner believes the average cost in Metro Z is not more than this standard .She takes a random sample of 20 households in Metro Z. Assume costs are approximately normal. Use $\alpha = 0.05$.

Sample data (Rs. 1,000):

1,185; 1,210; 1,195; 1,205; 1,180; 1,220; 1,190; 1,200; 1,175; 1,215; 1,188; 1,202; 1,193; 1,207; 1,199; 1,206; 1,182; 1,214; 1,191; 1,204

- a) State and perform an at most one-sided test at $\alpha = 0.05$ comparing the Metro Z mean to the standard.
- b) Construct a 95% confidence interval for the Metro Z mean and explain whether the CI supports the test conclusion.

Question# 13

A company wants to know whether the average cost per employee differs between its in-house and outsourced training programs.

Training Type	n	Mean Cost (Rs.)	Std. Dev (Rs.)
In-house	15	42,000	3,500
Outsourced	10	46,500	2,000

Assume costs are approximately normal and population variances are not equal.

- a) Test at $\alpha = 0.05$ whether there is a significant difference in the average cost per employee between the two programs.
- b) Construct a 95% confidence interval for difference in means and interpret the results.

Question#14

Two competing firms spend heavily on advertising. Management wants to check if Firm A's average monthly marketing expenditure is at most that of Firm B. The following summary statistics were obtained:

Firm	n	Mean (Rs. million)	Std. Dev (Rs. million)
A	9	12.3	2.1
B	11	13.5	1.4

Assume normal distributions but unequal population variances.

- a) State and test the hypotheses $H_0: \mu_A = \mu_B$, $H_1: \mu_A < \mu_B$ at $\alpha = 0.01$.
- b) Construct a 99% confidence interval for the difference and interpret.

Question#15

A financial analyst believes that the average monthly salary of employees in Bank X is at most that of Bank Y.

The data below summarize independent random samples:

Bank	n	Mean Salary (Rs.)	Std. Dev (Rs.)
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X	12	73,000	6,200
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Y	10	70,500	4,100
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Assume salaries are normally distributed, but population variances are unequal.

- Formulate and test the hypothesis that Bank X is no more than Bank Y
- Construct a 90% confidence interval for $\mu_X - \mu_Y$ and comment on whether it supports the test conclusion.

Question#16

A researcher wants to compare the average annual living cost (in Rs. 000) between City A and City B. Random samples of 10 families in each city produced the following results:

City	n	Mean (Rs. 000)	Std. Dev (Rs. 000)
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A	10	1850	75
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B	10	1780	45
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Assume costs are normally distributed but population variances are unequal.

- Using $\alpha = 0.05$, test whether there is a significant difference in the mean annual living cost between the two cities.
- Construct a 95% confidence interval for the difference in population means, $\mu_A - \mu_B$, and interpret your result.

Question#17

A consumer goods company suspects that the proportion of customers who recall seeing its TV ad differs between Urban and Rural markets. In a survey, Urban: 210 of 600 respondents recalled the ad; Rural: 138 of 520 respondents recalled it.

- Test at the 1% significance level whether there is a difference in recall proportions between Urban and Rural customers.
- Construct a 99% confidence interval for the difference
- Explain what the interval and test tell the company about regional ad effectiveness and possible next steps.

Question#17

A fintech firm ran two different landing-page designs to attract signups for a new app. From random visitors, Design A produced 92 signups out of 1,200 visitors, and Design B produced 75 signups out of 1,050 visitors.

- a) Test at the 5% significance level whether Design A has a higher conversion proportion than Design B.
- b) Construct a 95% confidence interval for the difference in proportions
- c) Interpret your results and advise which design the marketing team should prefer.

Question 18 — Employee Productivity (Before vs After Training)

A company wants to know whether a **new training program** improves employee productivity (units produced per day). The productivity of 10 randomly selected employees was measured **before** and **after** training:

Employee Before After

1	45	50
2	48	53
3	46	51
4	44	48
5	47	52
6	49	55
7	43	46
8	46	49
9	44	48
10	45	50

- a) Test at $\alpha = 0.05$ whether the training significantly increased productivity.
- b) Construct a **95% confidence interval** for the **mean improvement** in productivity.

Question 19— Sales Performance (Before vs After Advertisement)

A retailer launched a **new advertising campaign** and wants to test if it led to higher average weekly sales (in Rs. '000). The following are the **weekly sales before and after** the campaign for 8 randomly chosen outlets:

Outlet Before After

1	225	240
2	210	215
3	195	210
4	205	220
5	200	215
6	230	235
7	215	230
8	190	200

- Using $\alpha = 0.01$, test if the new advertisement **increased average sales**.
- Construct a **99% confidence interval** for the mean change in weekly sales.

Question 20— Customer Satisfaction (Service Redesign)

A hotel introduced a **new digital check-in system** to improve customer satisfaction. Ten returning customers rated their check-in experience **before** and **after** the change (scale: 1–10):

Customer Before After

1	6	8
2	5	7
3	7	8
4	6	9

Customer Before After

5	5	7
6	6	8
7	7	8
8	4	6
9	6	8
10	5	7

- a) Test at $\alpha = 0.05$ if the new system significantly improved satisfaction.
b) Construct a **95% confidence interval** for the mean change in satisfaction scores.

Question 21— Cost Reduction (Before vs After Automation)

A manufacturing firm automated part of its process to reduce **monthly production costs** (Rs. '000).

The same 12 units were studied before and after automation:

Unit Before After

1	115	108
2	118	110
3	120	114
4	122	116
5	119	112
6	125	118
7	121	114
8	117	109
9	124	117

Unit Before After

10 116 110

11 118 111

12 123 115

- a) At $\alpha = 0.05$, test whether the automation significantly **reduced average monthly cost**.
b) Construct a **95% confidence interval** for the **mean cost reduction**.